

篝火的声音

Campfire Sound

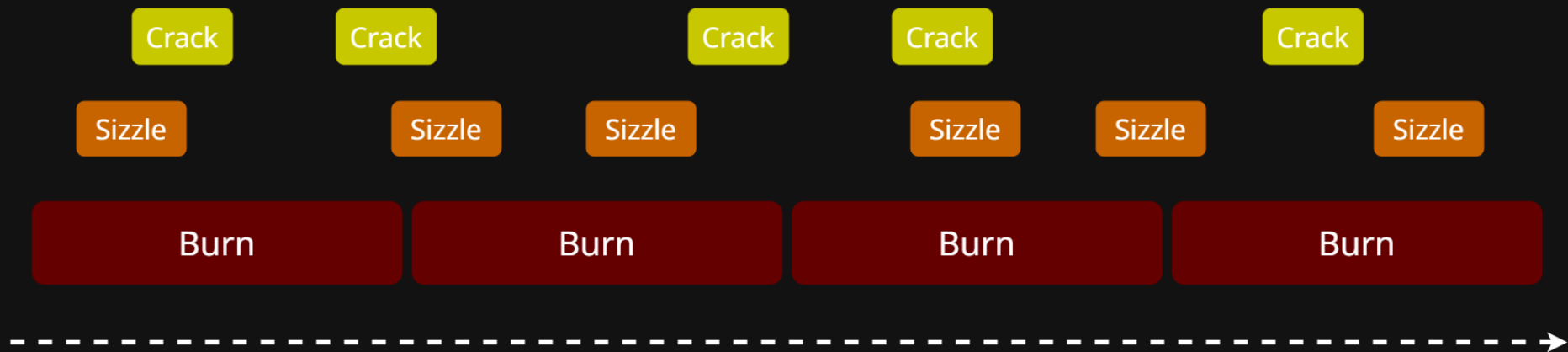
如何拆解篝火的声音

- 篝火燃烧时持续稳定的声音 Burn
- 零星火苗的声音 Sizzle
- 木头碎裂的声音 Crack



声音资源的规格

循环和随机



循环和随机

Loop and Randomize

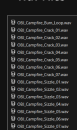
生成声音资产

SoundBank Generation

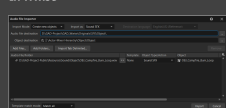
Wwise SoundBank Generation

Wav Files --> Wwise --> UE

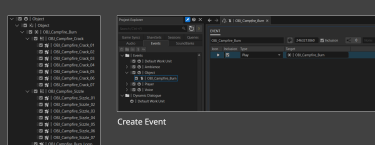
Wav Files



In Wwise

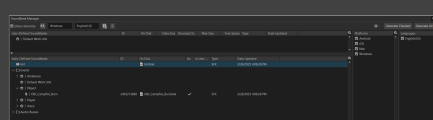


Import Audio Files into Wwise



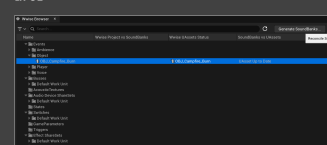
Create Event

Playback Logic with Containers

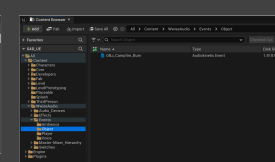


Generate SoundBank with Per Platform Settings

In UE

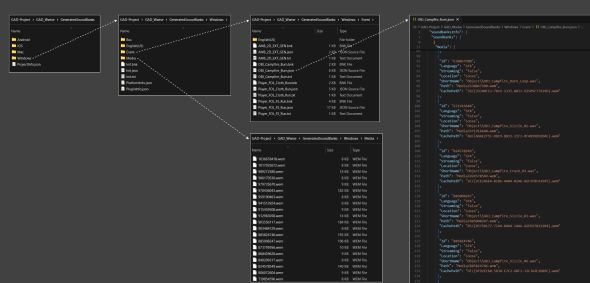


Reconcile SoundBank with Wwise Browser



AK Event Uasset Created

Under The Hood



音频空间化

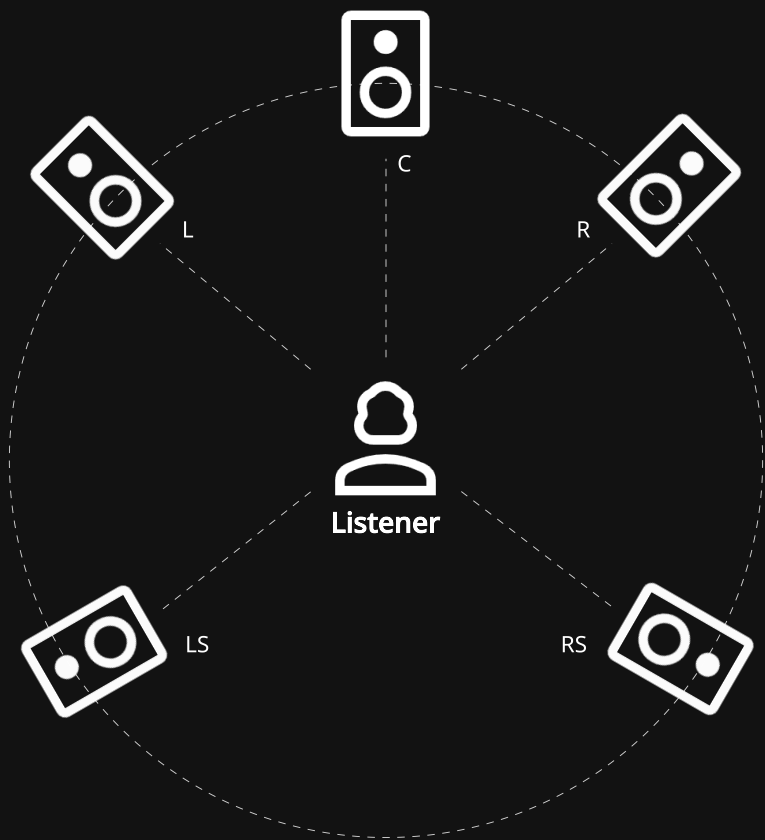
Audio Spatialization

还原真实世界中的声音传播效果

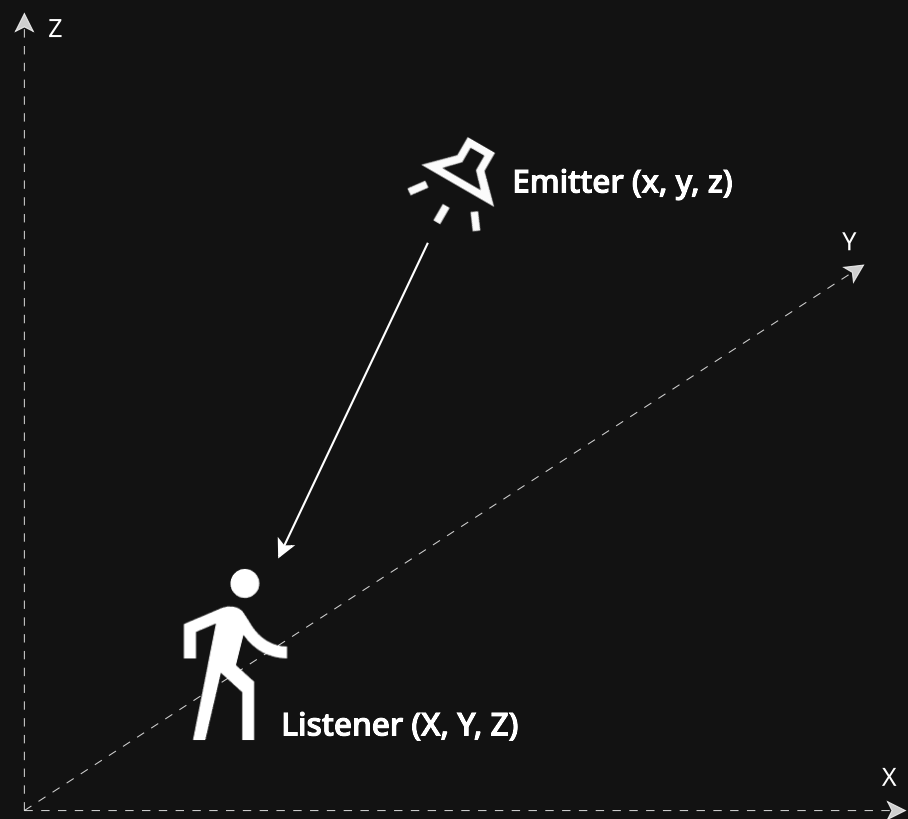
- 声像定位
- 空间划分
- 反射与混响
- 衍射与透射
- 声音衰减

声像定位 Sound Positioning

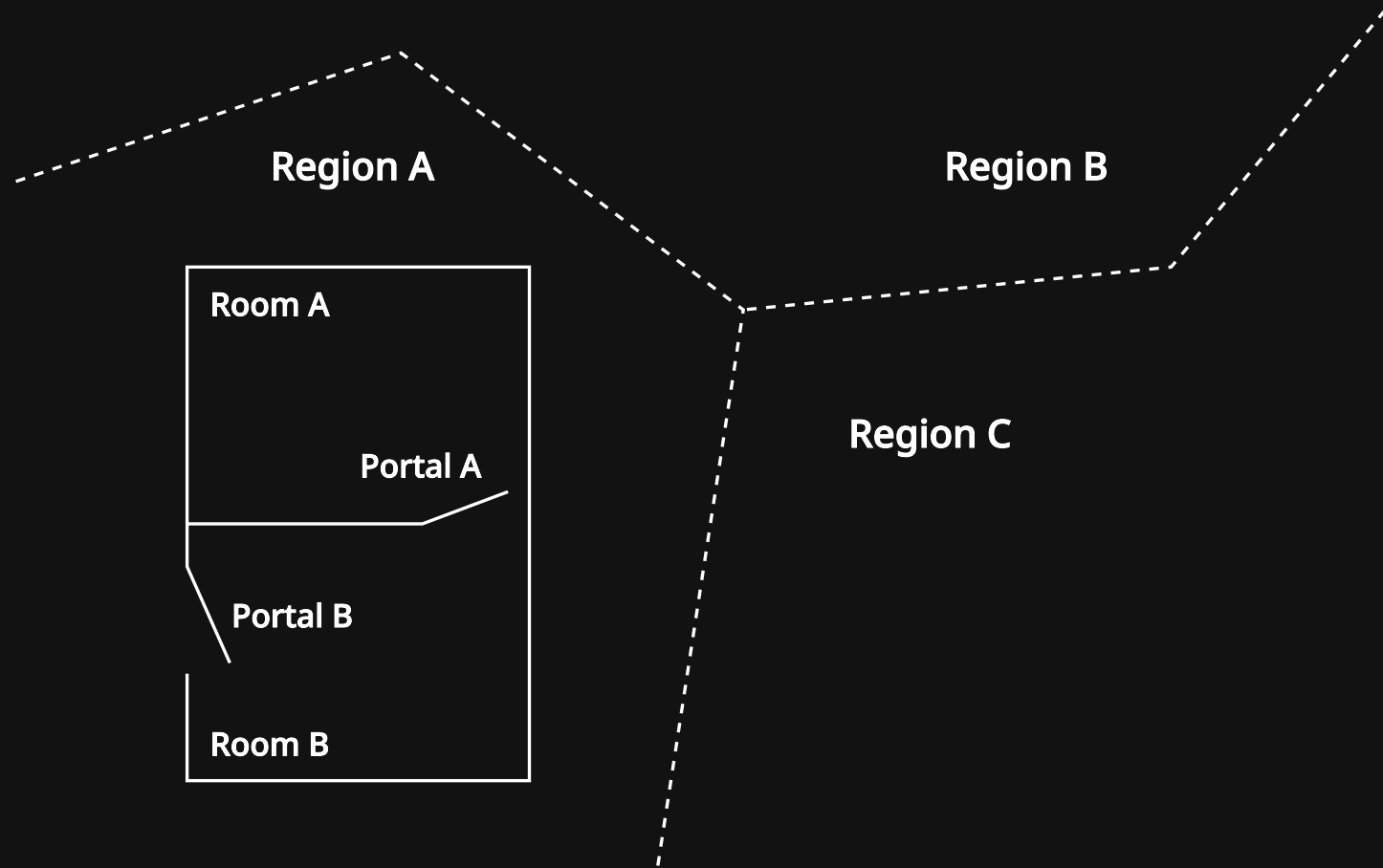
Channel-Based



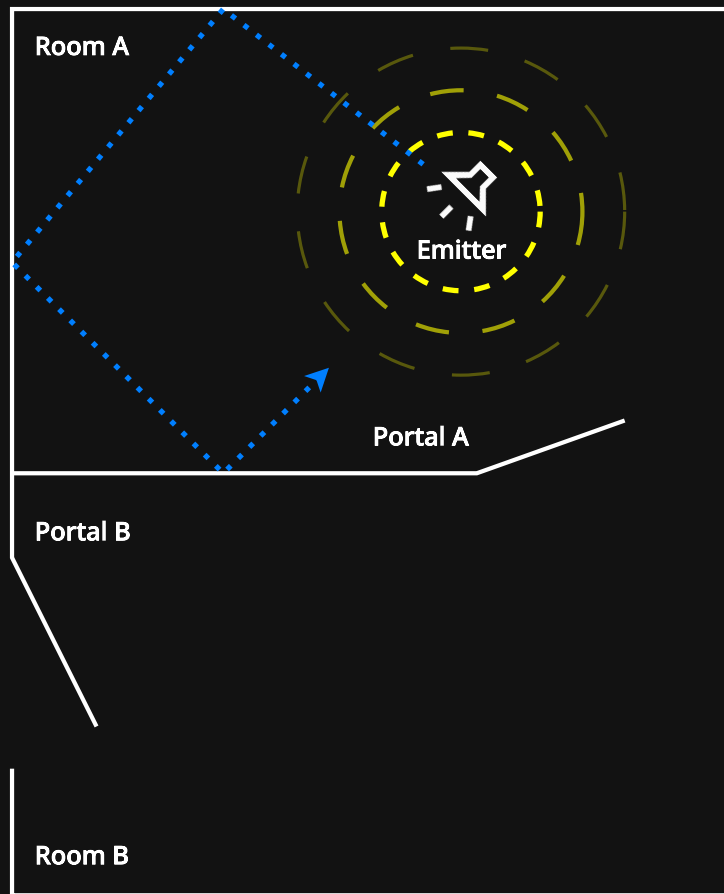
Object-Based



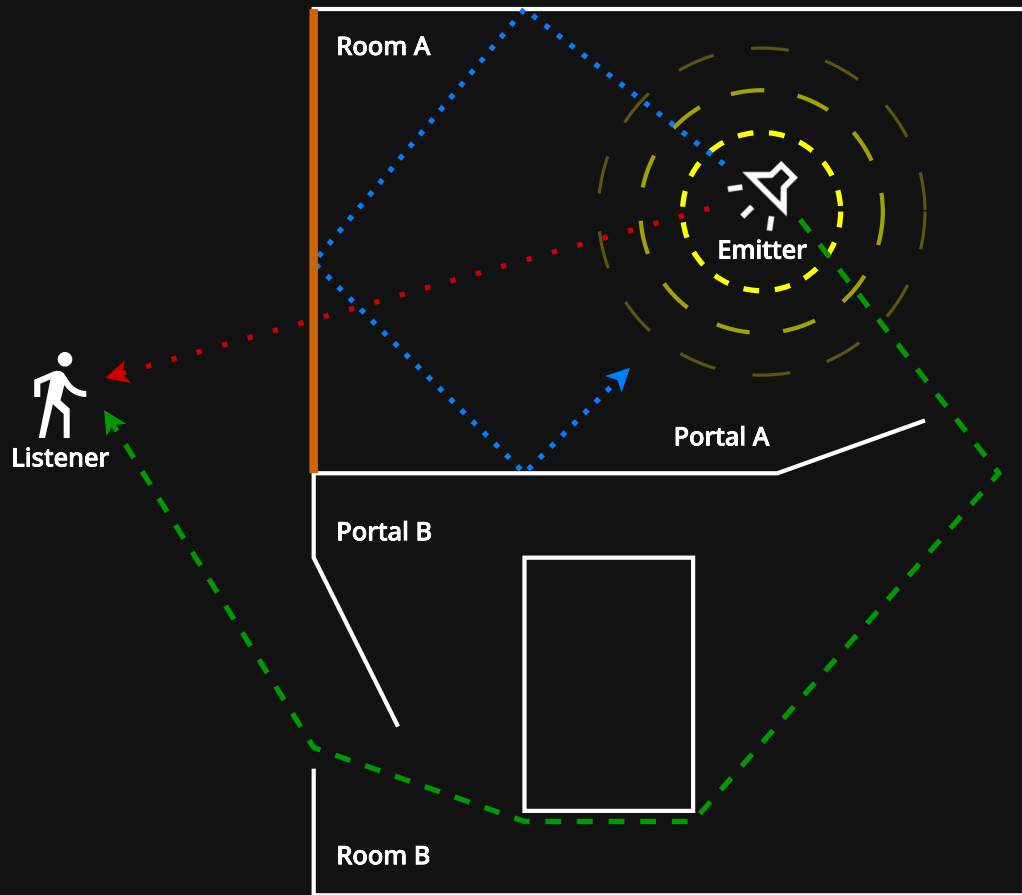
空间划分 Space Partition



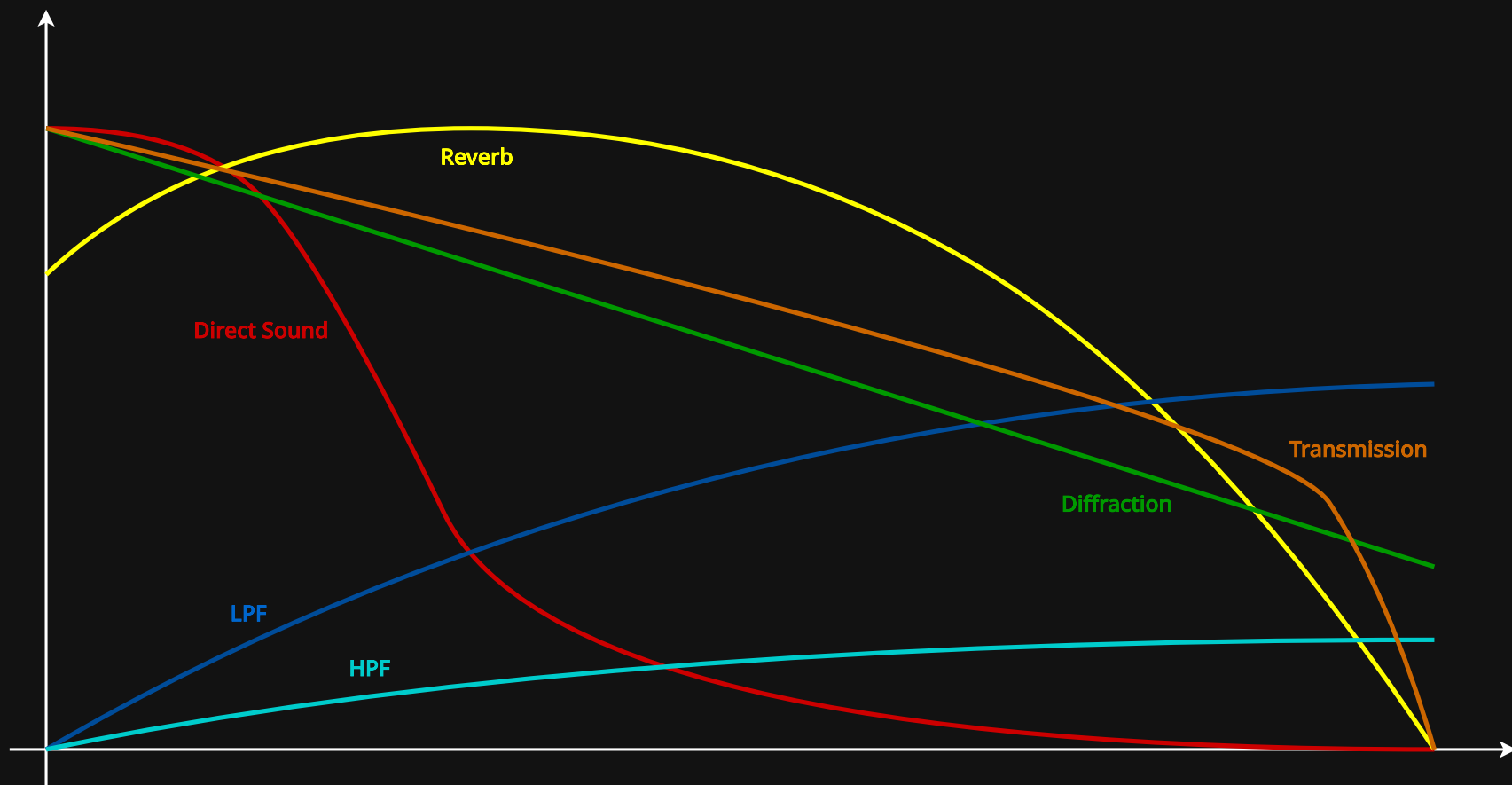
反射与混响 Reflect & Reverb



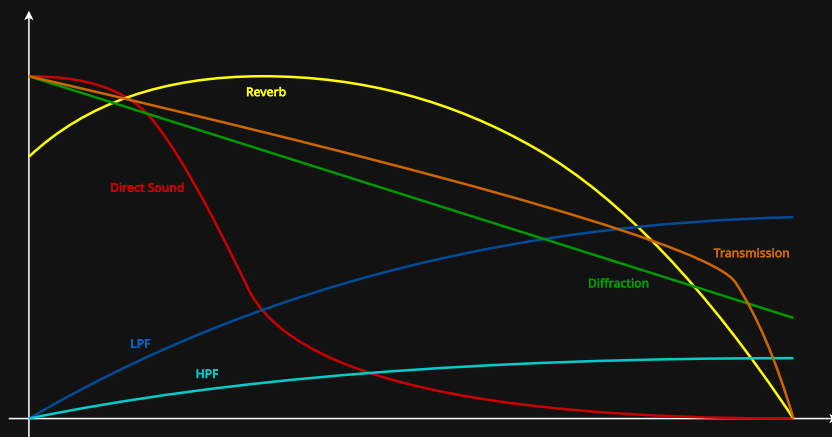
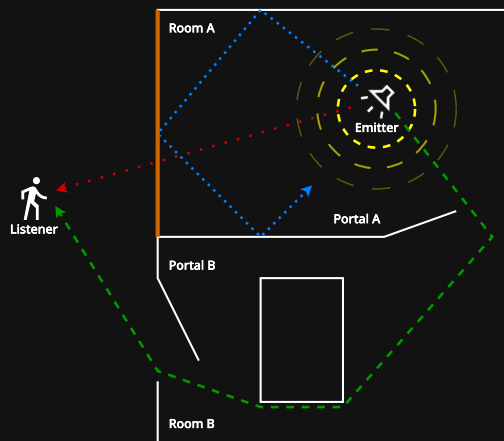
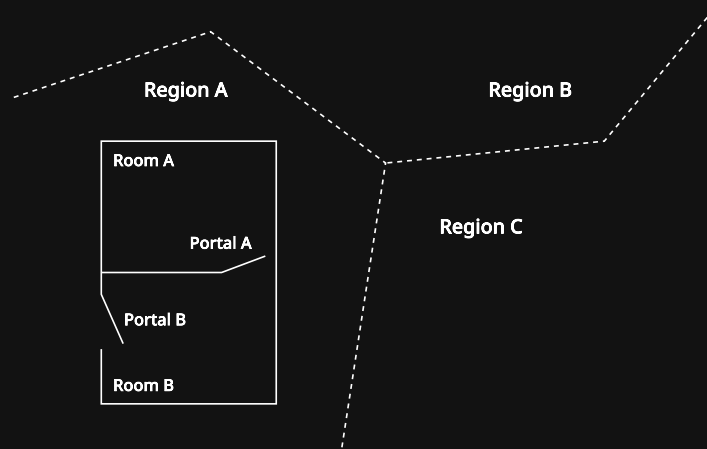
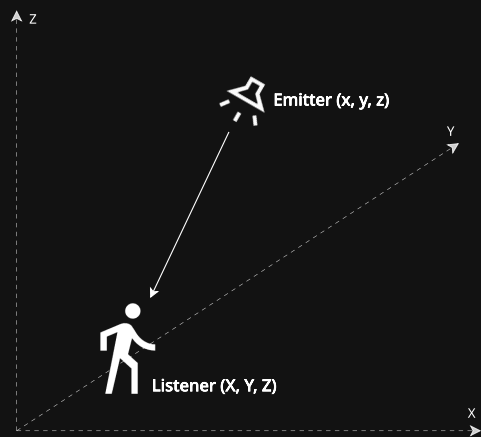
衍射与透射 Diffraction & Transmission



声音衰减 Sound Attenuation



音频空间化

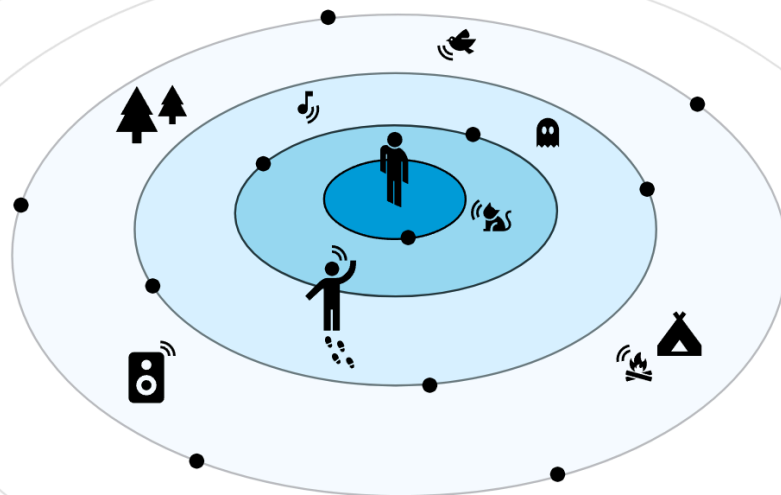


现成的解决方案

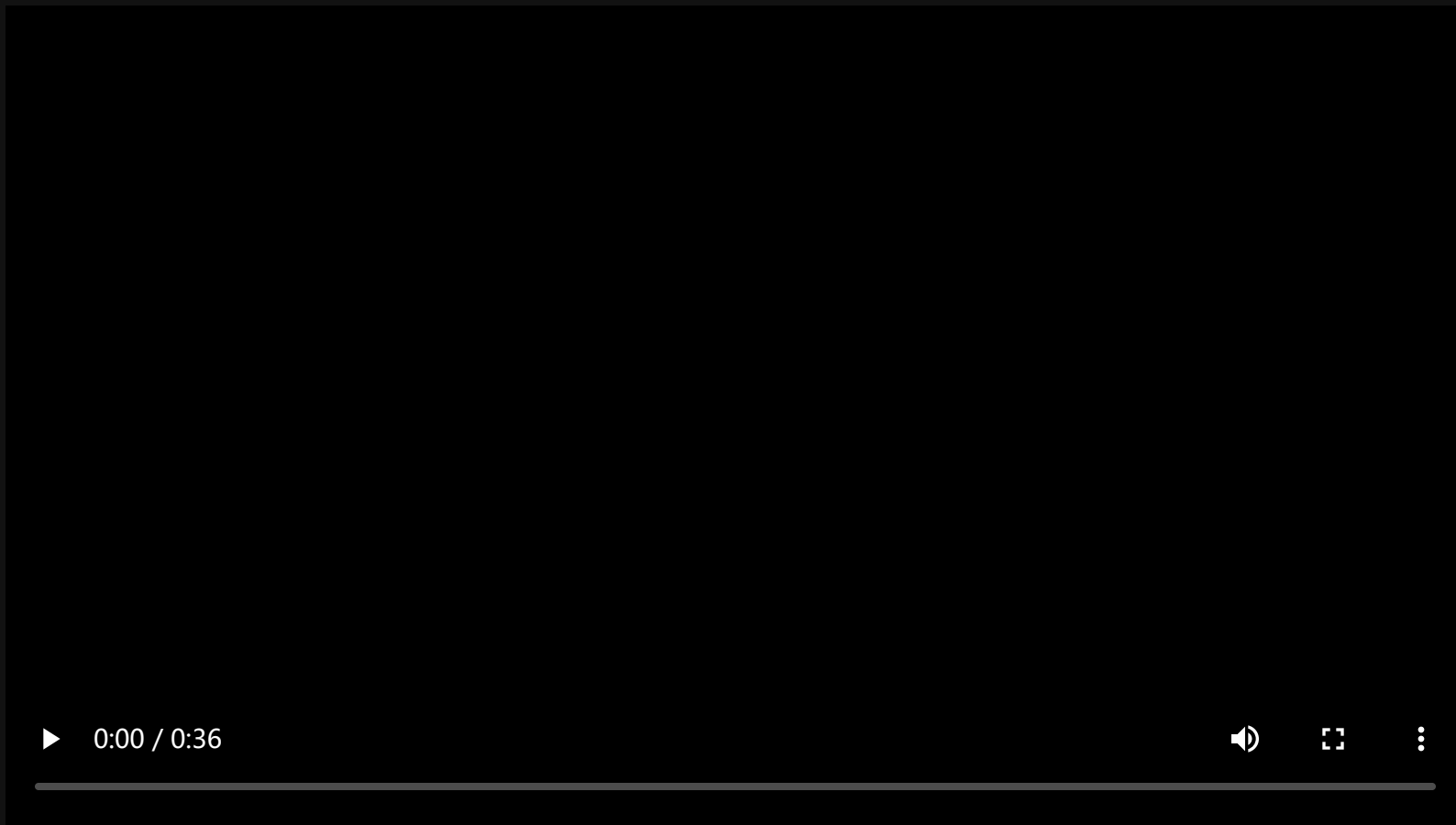
Wwise Spatial Audio

What is Wwise Spatial Audio?

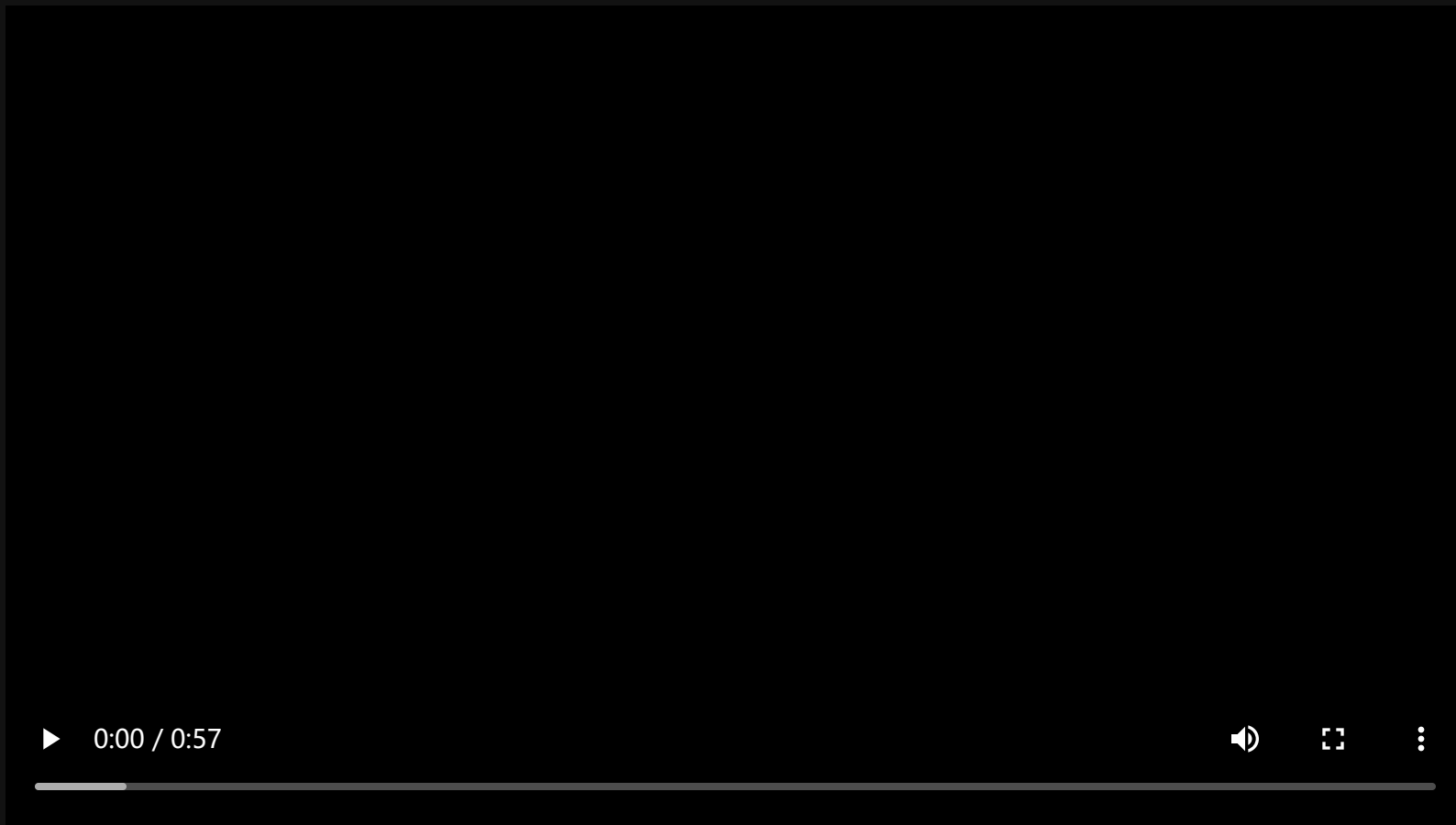
Wwise Spatial Audio is a family of built-in features complemented by in-house and 3rd-party plug-ins that seamlessly integrates with the Wwise production workflow. Wwise Spatial Audio encompasses sound propagation, virtual acoustics, and spatially informed audio rendering. It refers to any technology that processes audio with the goal of convincing you that the sounds originate from the world around you.



Wwise Spatial Audio Demo



Wwise Spatial Audio Demo



音频空间化是还原和创造数字声学空间的底层框架。

Audio spatialization is the fundamental framework for recreating and designing digital acoustic spaces.

环境声设计

Ambience Sound Design

这个场景包含了哪些声音元素

- 森林中的风
- 树叶摇晃
- 树枝断裂
- 溪流
- 篝火
- 夜晚虫鸣
- 夜行鸟类
- 其他生物
-

需要考虑哪些因素

- 世界观
 - 时代背景
 - 地理区域
 - 类型与风格
 - ...
- 声音种类
 - 风
 - 水
 - 植被
 - 生物
 -
- 资产规格
 - 可循环的 / 单次触发的
 - 声道数目
 - Quad
 - Stereo
 - Mono
 - 2D / 3D 声源
 - 点声源
 - 线声源
 - 体声源
 - 随机声源
 -
- 其他
 - 风的强度和方向
 - 水流缓急
 - 昼夜交替
 - 平面和高度上的考量
 -

规划声音资产

Name	Quantity	Duration	Asset Type	Emitter Type
AMB_Forest_2D_Wind_Gentle	1	30s	Quad / Loop	2D
AMB_Forest_2D_Insect_Group_Night	1	30s	Quad / Loop	2D
AMB_Forest_3D_Tree_Rustle	9		Stereo / One-Shot	3D_Scatter
AMB_Forest_3D_Tree_Crack	9		Mono / One-Shot	3D_Scatter
AMB_Forest_3D_Water_Stream	1	15s	Stereo / Loop	3D_Spot
AMB_Forest_3D_Insect_Cricket	7		Mono / One-Shot	3D_Scatter
AMB_Forest_3D_Insect_Grasshopper	5		Mono / One-Shot	3D_Scatter
AMB_Forest_3D_Bird_Owl_Hoot	5		Stereo / One-Shot	3D_Scatter
AMB_Forest_3D_Wolf_Howl	5		Stereo / One-Shot	3D_Scatter